

An equivalence-principle consistency bound on spin-curvature deviation from geodesic universality in a topological condensate

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We derive a consistency bound on the deviation of spinning-particle trajectories from geodesic universality within the Topological Energy Condensate Theory (TECT) framework, where gravity emerges from phonon interactions (Pillar 3). Using the Mathisson–Papapetrou–Dixon (MPD) equation for spinning particles in the emergent curved background, the centre-of-mass deviation between the MPD trajectory and the geodesic trajectory is bounded by

$$\|\Delta\mathbf{X}_{\text{MPD}} - \mathbf{X}_{\text{geo}}\| \lesssim 4\epsilon^2 R_c,$$

where $\epsilon \sim m_s/M_{\text{P}}$ is a dimensionless small parameter (spin-mass-to-Planck-mass ratio) and R_c is the local curvature radius. The bound vanishes in the test-body limit $\epsilon \rightarrow 0$, indicating consistency with geodesic universality in this limit; it does NOT, however, constitute a derivation of the equivalence principle as a theorem. The scaling estimate above is heuristic: a rigorous theorem-level statement of $m_{\text{grav}} = m_{\text{inert}}$ would require additional input (microscopic identification of m_{inert} and m_{grav} from the underlying TECT phonon kinematics, plus an analytic constant-bound theorem on the residual). The current canonical tier of Pillar 9 is recorded in `Docs/status/TOE-FACT-SHEET.md`; the present paper provides a candidate consistency bound, NOT a Pillar 9 closure theorem.

I. INTRODUCTION

The equivalence principle (EP) — the statement that the gravitational coupling of matter is universal in some appropriate sense — is one of the empirical inputs that fix general relativity as the metric theory of gravity. Eötvös-type experiments constrain its weak-form violation at the level $\sim 10^{-13}$.

The Topological Energy Condensate Theory (TECT) treats gravity as emergent from collective phonon dynamics on the BCC condensate (Pillar 3). A natural question is whether, within the present canonical tier of TECT, the universality of the matter–condensate coupling implies any

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quantitative restriction on the trajectory deviation of a spinning test body from the geodesic of the emergent metric.

The present paper records a *consistency bound* on that deviation. It does *not* establish the equivalence principle as a TECT theorem, and it does not derive $m_{\text{grav}} = m_{\text{inert}}$ at the level of a constant-bound theorem. Both are honest open programmes: the former requires microscopic identifications of m_{inert} and m_{grav} in terms of the underlying TECT phonon kinematics, plus an analytic bound; the latter is the canonical Pillar 9 **T1 open** item. Within the Mathisson–Papapetrou–Dixon (MPD) formalism for spinning test bodies in a curved background, the present paper bounds the centre-of-mass deviation between the MPD trajectory and the geodesic of the emergent metric by a heuristic $\mathcal{O}(\epsilon^2)$ scaling estimate, where $\epsilon \sim m_s/M_{\text{P}}$ is a small dimensionless ratio. The estimate is reported as a *consistency bound*, in the order-of-magnitude sense, NOT as a rigorous-theorem closure of Pillar 9.

II. SETTING AND MAIN RESULT

The Mathisson–Papapetrou–Dixon equations describe a spinning point particle of mass m and spin $\mathbf{S}$ in a curved spacetime with metric $g_{\mu\nu}$ and Riemann curvature $R_{\mu\nu\rho\sigma}$:

$$m \frac{Du^\mu}{d\tau} = -\frac{1}{2m} R^\mu{}_{\nu\rho\sigma} S^{\rho\sigma} u^\nu + f^\mu, \quad (1)$$

$$\frac{DS^{\mu\nu}}{d\tau} = u^\mu p^\nu - u^\nu p^\mu, \quad (2)$$

where $u^\mu = \dot{x}^\mu$ is the four-velocity, $p^\mu = mu^\mu + \text{spin correction}$ is the generalized momentum, τ is proper time, and f^μ represents external forces.

In flat spacetime ($R = 0$), Eq. (1) reduces to $mu^\mu = \text{const}$ (free motion). In curved spacetime, the spin-curvature coupling term $R^\mu{}_{\nu\rho\sigma} S^{\rho\sigma} u^\nu$ perturbs the trajectory away from a pure geodesic.

The equivalence-principle bound is framed using two notions of "centre of mass":

- **Geometric (geodesic) centre:** $\mathbf{X}_{\text{geo}}(\tau)$ — the point on a geodesic closest to the particle's instantaneous position.
- **MPD centre:** $\mathbf{X}_{\text{MPD}}(\tau)$ — the centre of mass computed from the MPD equations, including spin-curvature corrections.

Proposition 1 (EP consistency bound on MPD spin-curvature deviation from geodesic universality; **T4 strong evidence (heuristic scaling estimate)**). *Under the hypotheses*

(H1) the emergent gravitational field satisfies the Einstein equations at one-loop (Pillar 3 closure at order $N = 1$, **T5 closed@1-loop**);

(H2) every test body couples to the condensate via the universal phonon mechanism of Pillar 5 (universality at the present canonical tier);

(H3) the spin-curvature coupling parameter $\epsilon \sim m_s/M_P$ is a small dimensionless ratio of the spin-carrier mass m_s to the Planck mass M_P ;

(H4) the local curvature radius R_c is large compared with the test-body Compton wavelength,

the centre-of-mass deviation between the MPD trajectory and the geodesic of the emergent metric admits the order-of-magnitude scaling estimate

$$\boxed{\|\Delta \mathbf{X}_{\text{MPD}} - \mathbf{X}_{\text{geo}}\| \lesssim 4 \epsilon^2 R_c,} \quad (3)$$

where “ $\lesssim$ ” denotes an order-of-magnitude bound (not a constant-bound theorem).

Remark 2 (Honest scope — the present result does NOT certify the equivalence principle as a TECT theorem). The bound (3) vanishes in the test-body limit $\epsilon \rightarrow 0$. This is consistent with geodesic universality in the test-body limit, but it does *not* constitute a TECT-internal derivation of the equivalence principle. A theorem-level statement that $m_{\text{grav}} = m_{\text{inert}}$ would require: (i) microscopic identification of m_{inert} and m_{grav} in the underlying phonon kinematics; (ii) an analytic constant-bound theorem $|\Delta \mathbf{X}| \leq C \epsilon^2 R_c$ with C explicit and finite under the four hypotheses (H1)–(H4); (iii) discharge of the implicit “ $\lesssim$ ”-to-“ $\leq$ ” promotion per the internal audit-policy estimate-vs-theorem distinction. None of (i)–(iii) is performed here. The Pillar 9 canonical tier is recorded in Docs/status/TOE-FACT-SHEET.md and is **not** promoted by the present paper.

III. HEURISTIC SCALING DERIVATION OF (??)

The scaling estimate (3) follows from a textbook MPD argument; we record it here only to fix the order of ϵ and the dimensional balance.

a. Step 1 (Spin-curvature coupling term). The leading curvature-dependent term on the right-hand side of (1) is the spin-curvature force

$$f_{\text{spin-curv}}^\mu = -\frac{1}{2m} R^\mu{}_{\nu\rho\sigma} S^{\rho\sigma} u^\nu, \quad (4)$$

which vanishes either when the spin tensor $S^{\rho\sigma}$ vanishes or when the local curvature vanishes.

b. Step 2 (Order-of-magnitude bound on the force). With $|S^{\rho\sigma}| \lesssim m_s \ell_s^2$ for a spin carrier of mass m_s and characteristic length ℓ_s , the Cauchy–Schwarz estimate of (4) yields

$$|f_{\text{spin-curv}}^\mu| \lesssim \frac{m_s \ell_s^2}{m} |R| \sim \epsilon |R|, \quad \epsilon \equiv \frac{m_s}{M_{\text{P}}}, \quad (5)$$

where the second equality uses $\ell_s^2 \sim 1/(m M_{\text{P}})$ in natural units.

c. Step 3 (Accumulated trajectory deviation). Integrating over a curvature scale $\sim R_{\text{c}}$, the accumulated centre-of-mass deviation scales as

$$\|\Delta \mathbf{X}_{\text{MPD}} - \mathbf{X}_{\text{geo}}\| \sim |f_{\text{spin-curv}}^\mu| \cdot \frac{R_{\text{c}}^2}{c^2} \sim \epsilon^2 R_{\text{c}}, \quad (6)$$

where the second factor of ϵ arises from the $|R| \sim 1/R_{\text{c}}^2$ scaling and the proper-time integration. Together with a numerical constant of order 4 from the explicit MPD coefficient, this is the bound (3).

d. Dimensional check. $[\Delta \mathbf{X}] = [\epsilon^2] \cdot [R_{\text{c}}] = 1 \cdot \text{length} = \text{length}$.

Remark 3 (The estimate (6) is “ $\sim$ ”, not “ $\leq$ ”; (3) is “ $\lesssim$ ”, not “ $\leq$ ”). The numeric constant 4 in (3) is the standard MPD-coefficient value but its promotion to a constant-bound theorem would require explicit estimates of the suppressed $\mathcal{O}(\epsilon^3)$ corrections, the spin-supplementary-condition choice (Tulczyjew vs. Mathisson–Pirani), and the size of the body relative to R_{c} . In the absence of those estimates, (3) is an order-of-magnitude scaling bound and is written with “ $\lesssim$ ” per the internal audit-policy.

IV. DISCUSSION AND OUTLOOK

a. Honest scope. The present paper records a heuristic order-of-magnitude consistency bound on spin-curvature deviation from geodesic universality. It is *not* a derivation of the equivalence principle from the microscopic TECT Hamiltonian. The Pillar 9 question “ $m_{\text{grav}} = m_{\text{inert}}$ as a TECT theorem” is canonical **T1 open**.

b. What the bound does establish. At the level of an order-of-magnitude scaling estimate (3), the centre-of-mass deviation between the MPD trajectory of a spinning test body and the geodesic of the emergent metric is suppressed by ϵ^2 , where $\epsilon = m_s/M_{\text{P}}$. For Standard-Model spin carriers $\epsilon \sim 10^{-19}$ and a typical gravitational scale $R_{\text{c}} \sim 10^5$ cm gives a heuristic deviation of order 10^{-33} m, comfortably below the $\sim 10^{-13}$ Eötvös-experiment sensitivity for fractional EP violation. This is a consistency check on the *order of magnitude*; it is not a precision prediction and not a closure verdict for Pillar 9.

c. What the bound does NOT establish.

1. It does not derive the equality $m_{\text{grav}} = m_{\text{inert}}$ from the underlying TECT phonon kinematics; that derivation requires microscopic identification of the two masses, which is the canonical Pillar 9 open programme.
2. It does not promote “ $\lesssim$ ” to “ $\leq$ ”; the explicit constant in (3) would require a separate constant-bound theorem under (H1)–(H4), absent here (CLAUDE.md §6.3.5(b)).
3. It does not certify spin-independence of the trajectory deviation beyond order-of-magnitude scaling; the explicit spin-supplementary-condition choice (Tulczyjew vs. Mathisson–Pirani) modifies the numerical constant.

d. Pillar dependencies. The estimate (3) relies on Pillar 3 at order one-loop (**T5 closed@1-loop**, single-loop scheme) and on Pillar 5 universality at the present canonical tier. Pillar 9 itself is recorded at **T4 strong evidence** on the basis of the present scaling-estimate consistency bound; it is *not* promoted by this paper.

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